A value repeated at its own distance: OEIS A209470

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Abstract

OEIS A209470 carries an empirical recurrence of order 33 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical. The entry's condition reaches only as far as the value itself, which is bounded by the alphabet, so a window of that many lines decides it; the admissible windows are the vertices of a finite digraph and the arrays counted are exactly the walks in it. Hence $a(n)$ is a walk count on $S = 40$ vertices and is C-finite, and whether the conjectured recurrence annihilates it is settled by a finite exact computation, with the Cayley–Hamilton theorem bounding the work.

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1 The conjecture

OEIS A209470 is "Number of nX2 0..2 arrays with every element value z a city block distance of exactly z from another element value z.". It has offset 1 and begins

$$2, 13, 93, 635, 4357, 27989, 175599, 1097149, \ \ldots$$

The entry states, as a conjecture contributed by its author and never marked settled:

> Empirical: a(n) = 7*a(n-1) -7*a(n-2) -4*a(n-3) +66*a(n-4) +304*a(n-5) +718*a(n-6) +1784*a(n-7) -1409*a(n-8) -3322*a(n-9) -13020*a(n-10) -15966*a(n-11) -18299*a(n-12) -14110*a(n-13) -25099*a(n-14) -14477*a(n-15) -28326*a(n-16) +8287*a(n-17) -13319*a(n-18) +13953*a(n-19) -8803*a(n-20) +22525*a(n-21) +4118*a(n-22) +2705*a(n-23) +3161*a(n-24) +4528*a(n-25) -280*a(n-26) -1198*a(n-27) -384*a(n-28) -1636*a(n-29) -200*a(n-30) +80*a(n-31) +32*a(n-32) +128*a(n-33)

As of the "Last modified" line on the live entry (Oct 03 2025 15:15:52, revision 9) this is still recorded as empirical, and nothing on the entry records it as proved.

2 The count is a walk count

The entry counts arrays with 2 columns over the alphabet $\{0, 1, 2\}$, distance measured as $|\Delta\text{row}| + |\Delta\text{column}|$, and asks that every element of value z must have another element of value z exactly z steps away. An element of value 0 is 0 steps from itself and so always has such a partner. That is not an interpretation imposed on the entry: it is the reading the entry's own published terms select, and reading "another" as strictly some other cell disagrees with them at the very first term.

The condition ranges over the whole array, but only as far as the value itself reaches: a cell of value z looks exactly z steps away, and z is at most 2. So nothing more than 2 rows away can matter, and a cell's verdict is settled once the row 2 beyond it has arrived.

Carry the last 2 rows together with one bit for each of their cells, recording whether that cell has already found its partner. A new row is paired against the rows it can still reach, sets the bits it completes, and retires the row that has now seen everything it will ever see: the walk continues if that row's bits say what the entry asks and stops otherwise. The rows still held when the array ends are judged by the end vector, which is where the boundary is honest — a partner off the edge of the array is not a partner.

Merging vertices with identical futures leaves $S = 40$. An admissible array is exactly a walk, so with M the adjacency matrix and ι, τ the vectors recording which states may begin and end an array,

$$a(n) \;=\; \iota^{\top} M^{n-1} \tau,$$

the exponent fixed by the entry's own shape and checked against its published terms. In particular a is C-finite.

3 The proof

Lemma 1. *Let $q(t) = t^{33} - \sum_i c_i t^{33-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^{\top} M^{j} q(M) \tau$ for the corresponding walk index j, so the recurrence holds from some point on if and only if $\iota^{\top} M^j q(M)\tau = 0$ for all large j, and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+33} - \sum_i c_i M^{j+33-i}$. For the bound, M^S is an integer combination of $I, M, \ldots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^{\top} M^j q(M)\tau$ obey the monic recurrence given by the characteristic polynomial of M; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. A constant factor in front of a divides out of the residual and changes nothing here. □

Theorem 2.

$$a(n) \;=\; 7\,a(n{-}1){-}7\,a(n{-}2){-}4\,a(n{-}3){+}66\,a(n{-}4){+}304\,a(n{-}5){+}718\,a(n{-}6){+}1784\,a(n{-}7){-}1409\,a(n{-}8){-}3322$$

for every $n > 33$.

Proof. The vector $w = q(M)\tau$ was formed by 33 matrix–vector products in exact integer arithmetic and $\iota^{\top} M^j w$ evaluated for $j = 0, 1, \ldots,$ again exactly; those integers vanish from the index corresponding to $n = 33 + 1$ onwards and the run continued until the working vector was identically zero, which settles every larger j at once. □

4 Verification

Four checks, each independent of the algebra above.

First, the model reproduces all 20 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry: the digraph is built by reading the entry's English, and a misreading gives different counts.

Second, the reading was pinned before the digraph was written, by a program that enumerates every array of the given shape one by one and measures the distance between every pair of cells directly. That program shares no code with the transfer matrix, so an error in one does not hide an error in the other.

Third, the conjectured recurrence was evaluated directly on the published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Fourth, the annihilation test of Lemma 1 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A209470.

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[4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.